
**ADDIS ABABA UNIVERSITY
INSTITUTE OF TECHNOLOGY
SCHOOL OF MECHANICAL AND INDUSTRIAL ENGINEERING**

Course Title	Strength of Materials II
Course Code	MEng3111
Academic Year	2022 G.C
Module Name	Mechanics of Materials
Module No.	11
Instructor	Sitotaw M.
Credits	3
Year / Semester	3RD Year / Semester I
Pre-requisites	Strength of Materials I, Applied Mathematics II

COURSE OBJECTIVES

The course enables students to:

- Analyze two-dimensional complex stress problems;
- Solve strength problems by using energy method;
- Calculate critical load and stress in columns;
- Calculate stress and deflections in curved beams, rings, discs, and cylinders;
- Analyze torsion of non-circular and thin-walled sections; and
- Apply elastic failure theories to investigate strength of various loading.

COURSE DESCRIPTION

Mechanics of Materials is a course designed to enable students basic tools for stress, strain and deformation analysis. Methods of determining the stresses, strains and deformations produced by applied loads are presented. Engineering design concepts are intergrated throughout the course.

COURSE OUTLINE

CHAPTER 1: COMPLEX STRESSES

Stresses on Oblique Planes

Material Subjected to Pure Shear

Material Subjected to Combined Direct and Shear Stresses

Principal Plane Inclination in terms of the Associated Principal Stresses *[1½ Week]*

Graphical Solution – Mohr’s Stress Circle

Three Dimensional Stresses – Graphical Representation.

CHAPTER 2: ENERGY METHODS

Strain Energy

Strain Energy – Tension or Compression

Strain Energy – Shear

Strain Energy – Bending

Strain Energy – Torsion

Strain Energy of a Three– Dimensional Principal Stress System *[1½ Week]*

Work Done due to Suddenly Applied Load

Castigliano’s Theorem for Deflection

Unit – Load Method

Application of Castigliano’s Theorem to Angular Movements

CHAPTER 3: STATICALLY INDETERMINATE STRUCTURES

Introduction

Statically indeterminate axial members

Statically indeterminate torsional members	<i>[1½ Week]</i>
Statically indeterminate beams	

CHAPTER 4: CURVED BEAMS

Curved Beams with Small Initial Curvature	
Curved Beams with Large Initial Curvature	<i>[1½ Week]</i>
Stress in Hooks	
Deflection of Curved Beams	

CHAPTER 5: STRUTS

Euler's Theory	
Equivalent Strut Length	
Comparison of Euler's Theory with Experimental Results	
Euler "Validity Limit"	
Rankine or Rankine – Gordon Formula	<i>[1½ Week]</i>
Struts with Initial Curvature	
Struts with Eccentric Load	
Laterally Loaded Struts	

CHAPTER 6: UNSYMMETRICAL BENDING

Inclined Bending of Beams Having Symmetrical Cross Sections	
Beams Having Non – Symmetrical Cross Sections	
Properties of Cross Section	
Moment of Inertia	<i>[1½ Week]</i>
Product of Inertia	
Mohr's Circle of Inertia	
Stress Determination of Beams Having Non-Symmetrical Cross Section	
Deflection	

CHAPTER 7: THIN AND THICK CYLINDERS

Thin Cylinders	
Difference in Treatment between Thin and Thick Cylinders – Basic Assumptions	
Development of the Lamé Theory	
Thick Cylinder	<i>[1½ Week]</i>
Compound Cylinders	

CHAPTER 8: RINGS, DISCS and CYLINDERS SUBJECTED TO ROTATION AND THERMAL GRADIENTS

Thin Rotating Ring or Cylinder	
Rotating Solid Disc	<i>[1 Week]</i>
Rotating Disc with a Central Hole	
Rotating Thick Cylinders of Solid Shafts	
Rotating Disc of Uniform Strength	
Combined Rotational and Thermal Stresses in Uniform Discs and Thick Cylinders	

CHAPTER 9: TORSION OF NON – CIRCULAR AND THIN – WALLED SECTION

Rectangular Section	
Narrow Rectangular Sections	
Thin – Walled Open Sections	

Thin – Walled Split Tube

[1 Week]

Other Solid (Non – Tubular) Shafts

Thin – Walled Closed Tubes of Non – Circular Sections (Bredt – Batho Theory)

Thin – Walled Cellular Sections

Membrane Analogy

CHAPTER 10: THEORIES OF ELASTIC FAILURE

Maximum Principal Stress Theory

Maximum Principal Strain Theory

Maximum Shear Stress Theory

Total Strain Energy Theory

[1 Week]

Distortion Strain Energy Theory

Modified Maximum Principal Stress Theory for Brittle Materials

REFERENCE

1. Ferdinand P. Beer, Jr., E. Russell Johnston, and John T. DeWolf, *Mechanics of Materials*, Jan 20, 2005
2. Popov, E.P., *Mechanics of Materials (SI Version)*, 1978. (Old but still a good one.)
3. Beer, F.P. and Johnston E. Russell, *Mechanics of Materials*, 2005.
4. Robert L. Mott, *Applied Strength of Materials*, 2001.
5. Hearn, E.S., *Mechanics of Materials*, Aug. 1997
6. Andrew Pytel and Ferdinand L. Singer, *Strength of Material*, 1987
7. Nash, W.A., *Strength of Materials* (Schaum's Outline Series), Jul 1, 1998

EVALUATION:

Assignments	20%
Quiz	10%
Intermediate Exam	30%
Final Exam	40%

GRADING SYSTEM

As per the university legislation.

ATTENDANCE & DUE DATES

Students must have a minimum attendance of 75% during lecture & tutor classes, laboratory work and/or field trips. Should students fail to do so, they will not be allowed to sit for final exam. On the other hand, assignments shall be submitted on specified dates given by the instructor or else it will not be accepted.

STATEMENT OF ACADEMIC DISHONESTY

Cheating on exam, quizzes and Assignments is unacceptable conduct. Academic sanction will be taken against all parties involved in the act of cheating.